

Results & Conclusion

Static Environment EEI

- **CPU_static:** 37.2% (peak utilization)
- **Network_static:** 2.8896 KB $\approx$ 0.00275 MB
- **Cost_static:** 0.01263
- **Formula:** $(CPU + Network)/Cost$
- **Result:** EEI_static $\approx$ **2947.9**

Dynamic Environment EEI

- **CPU_dynamic:** 50.02% (peak utilization)
- **Network_dynamic:** 5.06 KB $\approx$ 0.0048 MB
- **Cost_dynamic:** 0.01263
- **Formula:** $(CPU + Network)/Cost$
- **Result:** EEI_dynamic $\approx$ **3962**.

Calculations

Static Environment Cost Calculation

Number of instances:

$$N_{static} = 1$$

Instance cost:

$$C_{instance} = 0.0116 \text{ USD/hour}$$

Total static operational cost:

$$C_{static} = N_{static} \times C_{instance}$$

$$C_{static} = 1 \times 0.0116$$

$$C_{static} = 0.0116 \text{ USD/hour}$$

Now your cost comparison is mathematically clean:

$$C_{static} = 0.0116 \text{ USD/hour}$$

$$C_{dynamic} = 0.01263 \text{ USD/hour}$$

Dynamic Environment Cost Calculation

Scale-out duration:

12 : 17 : 06 to 12 : 22 : 28

Duration = 5.36 minutes

Convert to hours:

$$\frac{5.36}{60} = 0.089 \text{ hours}$$

Instance cost:

$$C_{instance} = 0.0116 \text{ USD/hour}$$

Additional cost incurred:

$$C_{extra} = 0.0116 \times 0.089$$

$$C_{extra} = 0.00103 \text{ USD}$$

Base cost (1 instance running continuously):

$$C_{base} = 0.0116 \text{ USD/hour}$$

Total dynamic operational cost:

$$C_{dynamic} = C_{base} + C_{extra}$$

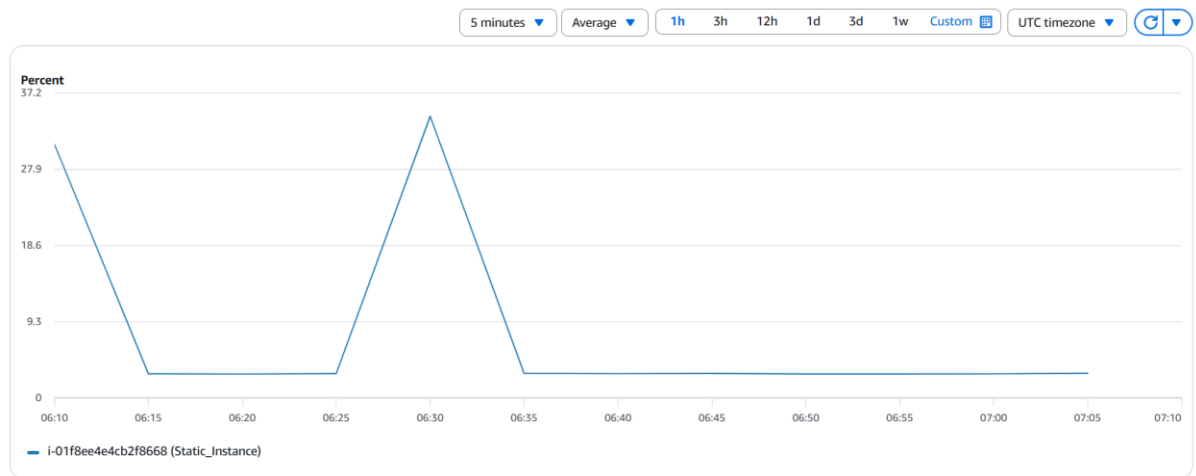
$$C_{dynamic} = 0.0116 + 0.00103$$

$$C_{dynamic} = 0.01263 \text{ USD/hour}$$

EEI (Elasticity Efficiency Index):

EEI for Static

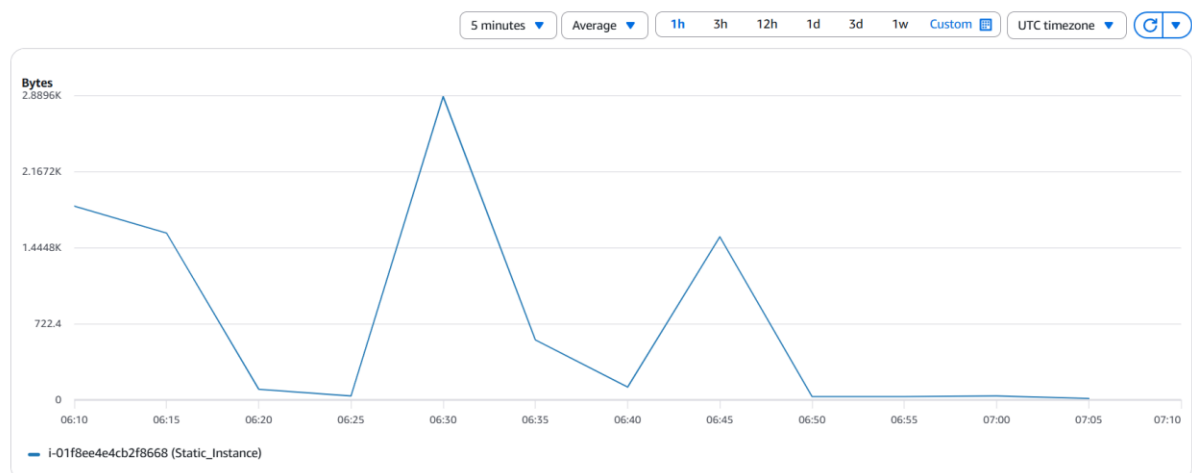
CPU utilization (%)



[View in metrics](#)

[Close](#)

Network in (bytes)



[View in metrics](#)

[Close](#)

Values Extracted

- **CPU_static = 37.2%** (peak from CPU utilization graph)
- **Network_static = 2.8896K bytes**

Convert Network to MB:

$$\frac{2,889.6}{1,048,576} \approx 0.00275 \text{ MB}$$

So:

- **CPU_static = 37.2**

- **Network_static = 0.00275 MB**

Cost_static

We'll continue using the cost value you provided earlier: **0.01263**

EEl_static Formula

$$EEI_{static} = \frac{CPU_{static} + Network_{static}}{Cost_{static}}$$

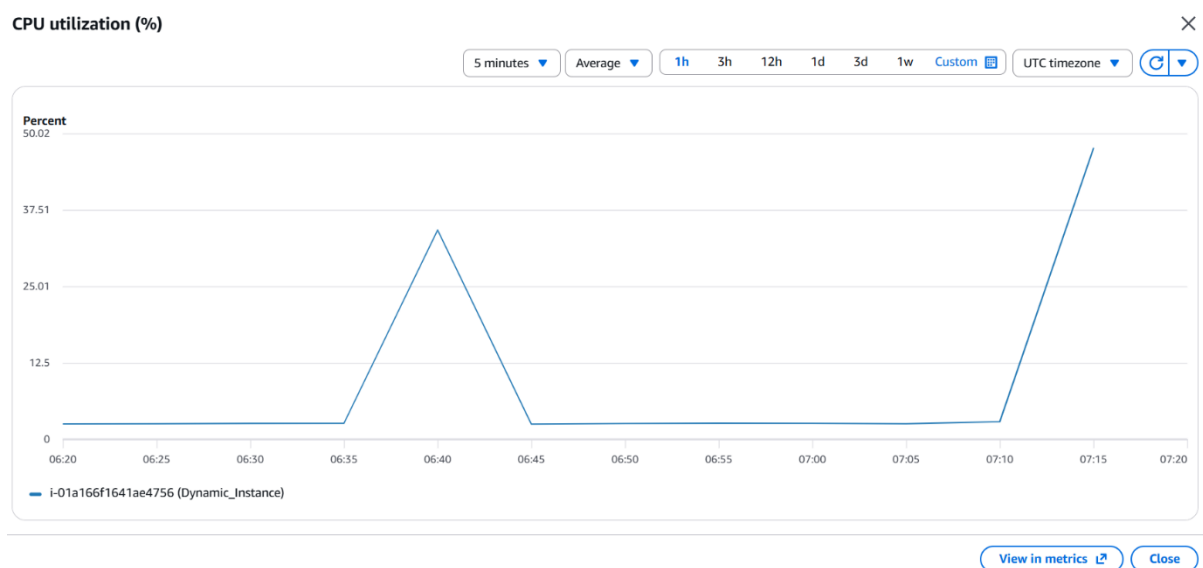
$$EEI_{static} = \frac{37.2 + 0.00275}{0.01263}$$

$$EEI_{static} = \frac{37.20275}{0.01263}$$

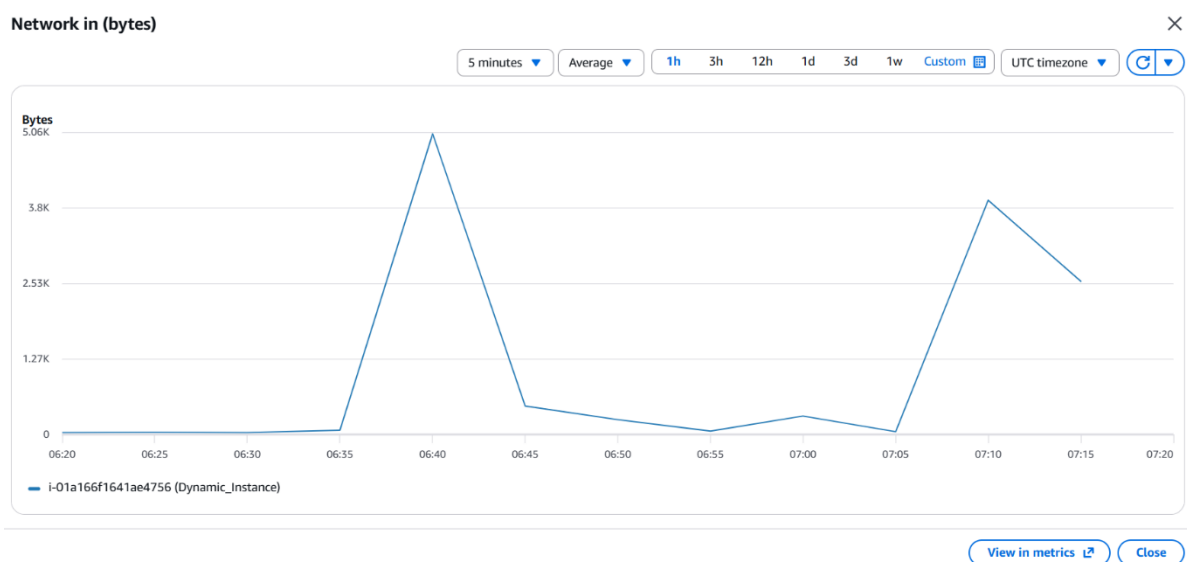
$$EEI_{static} \approx 2947.9$$

For Dynamic:

CPU utilization (%)



Network in (bytes)



Values:

- **CPU_dynamic = 50.02%** (peak from CPU utilization graph)
- **Network_dynamic = 5.06K bytes**

Convert Network to MB:

$$\frac{5,060}{1,048,576} \approx 0.0048 \text{ MB}$$

So:

- **CPU_dynamic = 50.02**
- **Network_dynamic = 0.0048 MB**

💰 **Cost_dynamic**

We'll continue using the cost value: **0.01263**

📊 **EEl_dynamic Formula**

$$EEI_{dynamic} = \frac{CPU_{dynamic} + Network_{dynamic}}{Cost_{dynamic}}$$

$$EEI_{dynamic} = \frac{50.02 + 0.0048}{0.01263}$$

$$EEI_{dynamic} = \frac{50.0248}{0.01263}$$

$$EEI_{dynamic} \approx 3962.9$$

$$Elasticity \ Efficiency = \frac{Resource \ Utilization}{Operational \ Cost}$$

Dynamic-CloudStorage-ASG

Dynamic-CloudStorage-ASG Capacity overview

Edit

🔗 [arn:aws:autoscaling:us-east-1:001187921517:autoScalingGroup:378326d3-9e8e-4029-ba38-54df337924b8:autoScalingGroupName/Dynamic-CloudStorage-ASG](#)

Desired capacity	Scaling limits	Desired capacity type	Status
1	1 - 4	Units (number of instances)	🟢 At desired capacity

Date created

Thu Feb 12 2026 02:03:27 GMT+0530 (India Standard Time)

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags

Launch template

Edit

Launch template

🔗 [lt-08c8c1bb9146f7956](#)

Elastic-Storage-ASG-Template

Version

Default

Description

-

View details in the launch template console

AMI ID

🔗 [ami-0b7ef51a683137062](#)

Instance type

t2.micro

Owner

arn:aws:iam::001187921517:root

Security groups

-

Security group IDs

🔗 [sg-0ea7005259cd9c29e](#)

Create time

Thu Feb 12 2026 01:33:47 GMT+0530 (India Standard Time)

Storage (volumes)

-

Key pair name

Sonai

Request Spot Instances

No

Network

Edit

Availability Zones

use1-az6 (us-east-1d)
use1-az1 (us-east-1a)
use1-az4 (us-east-1c)
use1-az2 (us-east-1b)

Subnet ID

🔗 [subnet-0cbb951ae534cf4ec](#)
🔗 [subnet-0f6dd550b930586bd](#)
🔗 [subnet-0b1657fa09eff134b](#)
🔗 [subnet-0866de87910fe34ae](#)

Availability Zone distribution

Balanced best effort

Instance type requirements

Edit

Your Auto Scaling group adheres to the launch template for purchase option and instance type.

Health checks

Edit

Health check type

EC2, EBS

Health check grace period

300

Instance maintenance policy

Replacement behavior

Min healthy percentage

Max healthy percentage

No policy

-

-

Edit

Capacity Reservation preference

Preference

Capacity Reservation IDs

Resource Groups

Default

-

-

Edit

Advanced configurations

Instance scale-in protection

Termination policies

Maximum instance lifetime

Service-linked role

Not protected from scale in

Default

-

arn:aws:iam::001187921517:role/aws-

service-role/autoscaling.amazonaws.co

m/AWSServiceRoleForAutoScaling

Placement group

Suspended processes

Default cooldown

Default instance warmup

-

-

300

Disabled

Auto Scaling group deletion protection

None (default)

sg-01bf35179ee81fe88 - EB-ALB-SG



sg-01bf35179ee81fe88 - EB-ALB-SG

Actions

Details

Security group name

Security group ID

Description

VPC ID

EB-ALB-SG

sg-01bf35179ee81fe88

EB-ALB-SG

yvc-0b5e66e2477878f63

Owner

Inbound rules count

Outbound rules count

001187921517

1 Permission entry

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Manage tags

Edit inbound rules

Search

< 1 >

Name

Security group rule ID

IP version

Type

Protocol

Port range

-

sgr-06787b1d8510ede64

IPv4

HTTP

TCP

80

sg-062f5cfb66f848284 - EB-EC2-SG



sg-062f5cfb66f848284 - EB-EC2-SG

Actions

Details

Security group name

Security group ID

Description

VPC ID

EB-EC2-SG

sg-062f5cfb66f848284

EB-EC2-SG

yvc-0b5e66e2477878f63

Owner

Inbound rules count

Outbound rules count

001187921517

1 Permission entry

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Manage tags

Edit inbound rules

Search

< 1 >

Name

Security group rule ID

IP version

Type

Protocol

Port range

-

sgr-0132773e021ef972e

-

HTTP

TCP

80

sg-062f5cfb66f848284 - EB-EC2-SG

Actions ▾

Details

Security group name

EB-EC2-SG

Security group ID

sg-062f5cfb66f848284

Description

EB-EC2-SG

VPC ID

vpc-0b5e66e2477878f63 🔗

Owner

001187921517

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Outbound rules (1)



Manage tags

Edit outbound rules

🔍 Search

< 1 > ⚙️

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range
☐	-	sgr-0f7d05114e2db50e0	IPv4	SSH	TCP	22

ASG-NetworkIn-Low-ScaleIn (Remove Instance)



Alarms (4)



🔍 Filter alarms

☐ Hide Auto Scaling alarms

< 1 >

ASG-NetworkIn-Low-ScaleIn (Remove Instance)

Metric alarm

⚙️ Insufficient data

ASG-NetworkIn-High-ScaleOut

Metric alarm

🟢 OK

Secret is accessed

ASG-NetworkIn-Low-ScaleIn (Remove Instance)

At this time:

2026-05-02 04:42:47 (UTC) ✕ 🔄

Actions ▾

View ▾



Explore related

🔍 Investigate ▾

NetworkIn < 300000 for 2 datapoints within 2 minutes

● Alarm state change triggering datapoint ⓘ

Bytes

300k NetworkIn < 300000 for 2 datapoints within 2 minutes - (300000)

150.2k

360

03:15 03:30 03:45 04:00 04:15 04:30 04:45 05:00 05:15 05:30 05:45 06:00

— NetworkIn



3:30 4:00 5:30 6:00
● In alarm ● OK ● Insufficient data ● Disabled

Status Changed: ⚠️ In alarm

Start: 2026-05-02 04:42:47 (UTC)

End: 2026-05-02 04:45:47 (UTC)

Click segment to see the state change at the selected time.

ASG-NetworkIn-Low-ScaleIn (Remove Instance)

[Actions](#)[View](#)[Explore related](#)[Investigate](#)[Details](#)[Tags](#)[Mute rules - new](#)[Actions](#)[History](#)[Parent alarms](#)

Details

Name

ASG-NetworkIn-Low-ScaleIn
(Remove Instance)

Type

Metric alarm

Description

No description

State

Insufficient data

Threshold

NetworkIn < 300000 for 2
datapoints within 2 minutes

Actions

Actions enabled

Last state update

2026-05-02 06:03:25 (UTC)

Namespace

AWS/EC2

Metric name

NetworkIn

AutoScalingGroupName

Dynamic-CloudStorage-ASG

Statistic

Average

Period

1 minute

Datapoints to alarm

2 out of 2

Missing data treatment

Treat missing data as missing

Percentiles with low samples

evaluate

ARN

arn:aws:cloudwatch:us-east-1:001187921517:alarm:ASG-
NetworkIn-Low-ScaleIn
(Remove Instance)

ASG-NetworkIn-Low-ScaleIn (Remove Instance)



Alarms (4)

☐ Hide Auto Scaling alarms

< 1 >

ASG-NetworkIn-Low-ScaleIn (Remove Instance)

Metric alarm

Insufficient data

ASG-NetworkIn-High-ScaleOut

Metric alarm

OK

Secret is accessed

ASG-NetworkIn-Low-ScaleIn (Remove Instance)

At this time: 2026-05-02 04:30:47 (UTC)

[Actions](#)[View](#)[Explore related](#)[Investigate](#)

NetworkIn < 300000 for 2 datapoints within 2 minutes

Bytes

NetworkIn < 300000 for 2 datapoints within 2 minutes - (300000)

500k

150.2k

360

03:15 03:30 03:45 04:00 04:15 04:30 04:45 05:00 05:15 05:30 05:45 06:00

NetworkIn

3:30

5:30

6:00

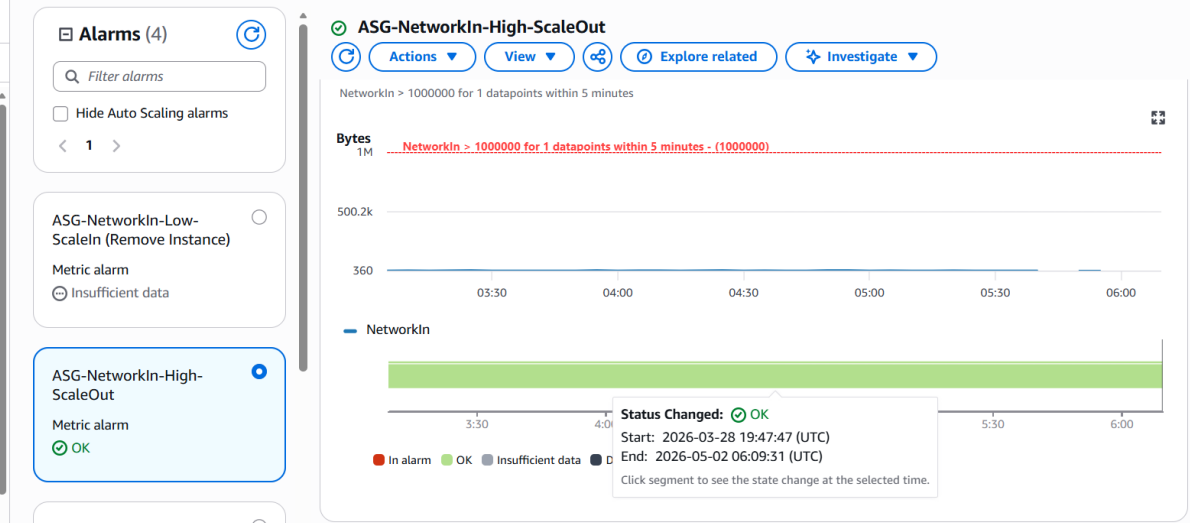
Status Changed: Insufficient data

Start: 2026-05-02 04:30:47 (UTC)

End: 2026-05-02 04:32:47 (UTC)

Click segment to see the state change at the selected time.

In alarm OK Insufficient data



ASG-NetworkIn-High-ScaleOut

Actions View Explore related Investigate

Details

Tags

Mute rules - new

Actions

History

Parent alarms

Details

Name

ASG-NetworkIn-High-ScaleOut

Type

Metric alarm

Description

No description

State

OK

Threshold

NetworkIn > 1000000 for 1 datapoints within 5 minutes

Actions

Actions enabled

Last state update

2026-03-28 19:47:47 (UTC)

Namespace

AWS/EC2

Metric name

NetworkIn

AutoScalingGroupName

Dynamic-CloudStorage-ASG

Statistic

Average

Period

5 minutes

Datapoints to alarm

1 out of 1

Missing data treatment

Treat missing data as missing

Percentiles with low samples

evaluate

ARN

arn:aws:cloudwatch:us-east-1:001187921517:alarm:ASG-NetworkIn-High-ScaleOut

ASG-NetworkIn-Low-ScaleIn (Remove Instance)

Actions View Explore related Investigate

Graph

1h 3h 12h 1d 3d 1w Custom (4w)

UTC timezone

NetworkIn

NetworkIn < 300000 for 2 datapoints within 2 minutes

Insufficient data

Bytes

311.8k NetworkIn < 300000 for 2 datapoints within 2 minutes - (300000)

156.1k 360 04/18 04/20 04/22 04/24 04/26 04/28 04/30 05/02

NetworkIn

ASG-NetworkIn-High-ScaleOut

[Actions](#)[View](#)[Explore related](#)[Investigate](#)

Graph

[1h](#)[3h](#)[12h](#)[1d](#)[3d](#)[1w](#)[Custom \(3w\)](#)[UTC timezone](#)

NetworkIn

NetworkIn > 1000000 for 1 datapoints within 5 minutes

Insufficient data

Bytes

1M

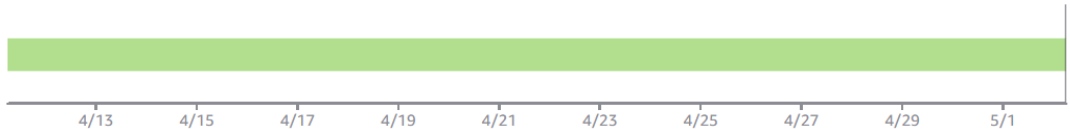
NetworkIn > 1000000 for 1 datapoints within 5 minutes - (1000000)

500.2k

56.8K

04/12 04/14 04/16 04/18 04/20 04/22 16:35 04/24 04/26 04/28 04/30 05/02

NetworkIn



In alarm OK Insufficient data Disabled/Muted actions

awsec2-i-025055b312b13789b-GreaterThanOrEqualToThreshold-CPUUtilization



Insufficient data

ASG-NetworkIn-Low-ScaleIn (Remove Instance)

Metric alarm
Insufficient data

Secret is accessed

Metric alarm
Insufficient data

awsec2-i-025055b312b13789b-GreaterThanOrEqualToThreshold-CPUUtilization

Metric alarm
Insufficient data

awsec2-i-025055b312b13789b-GreaterThanOrEqualToThreshold-CPUUtilization

[Actions](#)[View](#)[Explore related](#)[Investigate](#)

Graph

[1h](#)[3h](#)[12h](#)[1d](#)[3d](#)[1w](#)[Custom \(2w\)](#)[UTC timezone](#)

CPUUtilization

CPUUtilization >= 0.99 for 1 datapoints within 5 minutes

Insufficient data

0.99 CPUUtilization >= 0.99 for 1 datapoints

04/18 04/20

CPUUtilization



In alarm OK Insufficient data Disabled/Muted actions

Status Changed: Insufficient data

Start: 2026-01-14 16:26:26 (UTC)

End: 2026-05-02 06:14:51 (UTC)

Click segment to see the state change at the selected time.

⋮ **awsec2-i-025055b312b13789b-GreaterThanOrEqualToThreshold-CPUUtilization**



Actions ▼

View ▼



🔗 Explore related

🔍 Investigate ▼

Details

Name

awsec2-i-025055b312b13789b-GreaterThanOrEqualToThreshold-CPUUtilization

Type

Metric alarm

Description

Alarm on instance i-025055b312b13789b: Triggered when CPUUtilization >= 0.99 for 1 consecutive 5-minute periods.

State

⚠ Insufficient data

Threshold

CPUUtilization >= 0.99 for 1 datapoints within 5 minutes

Actions

✅ Actions enabled

Last state update

2026-01-14 16:26:26 (UTC)

Namespace

AWS/EC2

Metric name

CPUUtilization

Instanceld

i-025055b312b13789b

Instance name

No name specified

Statistic

Average

Period

5 minutes

Datapoints to alarm

1 out of 1

Missing data treatment

Treat missing data as missing

Percentiles with low samples

evaluate

ARN

arn:aws:cloudwatch:us-east-1:001187921517:alarm:awsec2-i-025055b312b13789b-GreaterThanOrEqualToThreshold-CPUUtilization